

AI / GenAI / RAG / MCP

Complete Job Roadmap

5 Courses | Step-by-Step Sections | 5 Projects

Telugu Learner Edition | 2026

Your Profile	Time Available	Expected Result
Basic Python + Job Goal Fresher / Career Change	1-4 hrs/day (Flexible)	5 Projects + GitHub RS 8-15 LPA Job Ready

COURSES OVERVIEW — What to Use & What to Skip

#	Course Name	Total	Watch	Skip	Priority
C1	AI Agent Dev (LangChain, Ollama, MCP)	5h 6m	~2.5h	~2.5h	HIGH
C2	Python Megaclass (300 Projects)	Very Long	~10h (Wk14,15,16)	90% Skip	MEDIUM
C3	Level Up LangChain + OpenAI	3h 51m	~3h	~45min	HIGH
C4	LangChain MasterClass (7 Apps)	~4h	~2.5h	~1.5h	MEDIUM
C5	Complete RAG Bootcamp	6h 31m	~4.5h	~2h	HIGH

STEP-BY-STEP LEARNING ORDER (Optimal Sequence)

STEP 1	Python Refresh	C2 — Week 14, 15, 16	~10 hrs
STEP 2	RAG Theory + Embeddings	C5 — Section 1, 3	~3.5 hrs
STEP 3	Prompt + Embeddings Deep	C4 — Section 1,2,3,4,7,8	~3 hrs
STEP 4	Full LangChain + RAG Build	C3 — Section 2,3,4,5,6,9,10,11	~3 hrs
STEP 5	MCP + Ollama + AI Agents	C1 — Section 1, 2, 3	~5 hrs
STEP 6	Advanced RAG + Agentic RAG	C5 — Section 4,5,7,9	~5.5 hrs
STEP 7	Memory + ChatGPT Clone	C4 — Section 10, 11	~3 hrs
STEP 8	Projects + GitHub + Deploy	All Courses — Build & Push	~10 hrs

COURSE 2 — Python Megaclass (Step 1: Python Refresh)

Course: 2025 AI Python Development Megaclass — 300 Hands-on Projects

Why This First? Basic Python ██████████ AI courses easy ██████████

Total Course: Very Long — █████ █████ 3 Weeks ██████████ █████!

Watch Only: Week 14, Week 15, Week 16 — ~10 hours

Week 14 — Python Basics Refresh (Watch 6 of 8 videos)

Video	Topic	Action	Why Important
106	Learn Python Quick Tutorial	YES	Complete overview
107	Hello World / Print	Not needed	Too basic
108	Variables & Data Types	YES	AI code ███ use ██████████
109	User Input & String Format	YES	CLI apps ███ ████████
110	If-Else Statements	YES	Logic building
111	Loops (For & While)	YES	Data iteration
112	Functions	YES	Code reuse
113	Lists & Shopping App	YES	Collections basic

Week 15 — Intermediate Python (Watch 4 of 7 videos)

Video	Topic	Action	Why Important
114	Dictionaries — Contact Book	YES	AI responses dictionary format ███ ██████████
115	Tuples & Sets	Skip	Job ███ immediately ██████████
116	File Handling — Note App	YES	JSON/PDF files read ██████████
117	Exception Handling	YES	Production code ███ must!
118	Return Values	Skip	Basic, ██████████
119	List Comprehensions	Skip	Later ████████████████████
120	Modules & Libraries	YES	pip install ████████████████████

Week 16 — Working with Data & APIs (Watch 4 of 7 videos)

Video	Topic	Action	Why Important
121	Reading Files	Skip	Week 15 ███ covered
122	Writing Files	Skip	Not urgent
123	CSV Files	YES	Data handling
124	JSON Files	YES	AI APIs ███████ JSON!
125	Weather API — Basics	YES	API calls ████████████████████
126	Dates & Time	Skip	Not urgent
127	Web Scraping	YES	Bonus skill!

Week 12 — Transformers Theory (2 Videos Only)

Video	Topic	Action	Why Important
091	Attention Mechanisms	YES	Interview ███ ████████████████████!
092	Transformer Architecture	YES	LLM basics ████████████████████

093-097	Deep Transformer Theory	Skip all	Time waste ████████
SKIP COMPLETELY: Weeks 1-11 (Math, ML, Deep Learning, CNNs, RNNs)			
SKIP COMPLETELY: Weeks 17-24 (GUI, Flask, OOP Projects, ML Algorithms)			
REASON: AI Agent job ███ immediately ██████████ — Time save ████████!			

COURSE 5 — Complete RAG Bootcamp (Step 2: RAG Theory)

Course:	Complete RAG Bootcamp — Build, Optimize, Deploy AI Apps
Published:	October 2025 — Fresh & Updated Content!
Total:	6h 31m — Watch ~4.5 hrs (skip ~2 hrs)
Why Now?	RAG = Most demanded skill in 2026 AI jobs

Section 1 — Introduction to RAG — WATCH ALL 3 Videos

Video	Topic	Action	Why Important
1.1	What is RAG	YES	Foundation concept
1.2	RAG System Overview	YES	Architecture clear ██████████
1.3	Applications of RAG	YES	Real-world use cases

Section 2 — RAG Architecture — Watch 1 of 3 only

Video	Topic	Action	Why Important
2.1	The Retrieval Process	Already in C3	
2.2	The Generation Process	Already in C3	
2.3	Putting It Together	YES	Complete picture ████████

Section 3 — Embeddings + Vector DB — WATCH ALL 3 Videos

Video	Topic	Action	Why Important
3.1	What Are Embeddings	YES	Core AI concept!
3.2	Intro to Vector Databases	YES	Pinecone vs Chroma compare
3.3	Building & Querying Vector Stores	YES	Hands-on practice

Section 4 — LangChain RAG Pipelines — WATCH ALL 3 Videos

Video	Topic	Action	Why Important
4.1	LangChain Core Concepts	YES	Reinforcement
4.2	RAG Implementation LangChain	YES	Deeper version
4.3	Adding Context & Metadata	YES	Advanced skill!

Section 5 — Enhancing RAG Performance — WATCH ALL 3 Videos

Video	Topic	Action	Why Important
5.1	Advanced Retrieval Techniques	YES	Re-ranking, MMR
5.2	Optimizing Context & Prompts	YES	Salary boost skill!
5.3	Evaluation and Metrics	YES	Interview must!

Section 6 — Deploying RAG — Watch 1 of 3 only

Video	Topic	Action	Why Important
6.1	Frontend Integration	Done in C3	
6.2	Backend APIs (FastAPI)	YES	FastAPI + RAG deploy
6.3	Deployment & Scalability	Done in C3	

Section 7 — Advanced + Hybrid RAG — WATCH ALL 3 Videos

Video	Topic	Action	Why Important
7.1	Hybrid Search (Keyword+Vector)	MUST	2026 hottest skill!
7.2	Multi-Modal RAG	YES	Images + Text RAG
7.3	Agentic RAG	MUST	Top job requirement!

Section 8 — Real World Cases — SKIP ENTIRE SECTION

Video	Topic	Action	Why Important
8.1-8.3	Enterprise, Security, Workflows	Theory heavy	After job ██████████

Section 9 — Capstone Project — WATCH ALL + BUILD

Video	Topic	Action	Why Important
9.1	RAG for Developers	YES	Career guidance
9.2	Capstone Project	+ BUILD	Your Portfolio Project 5!

COURSE 4 — LangChain MasterClass (Step 3: Prompts & Memory)

Course:	LangChain MasterClass — Develop 7 OpenAI LLM Apps
Warning:	2023 ■■■ ■■■■■■■■ — Concepts ■■■■■■■■, exact code ■■■■■
Watch:	Section 1,2,3,4,7,8,10,11 — ~6 hours
Skip:	Projects 1,2,4 — Already done better in other courses

Sections 1+2 — LangChain & OpenAI Intro — WATCH ALL

Video	Topic	Action	Why Important
S1-V1	What is LangChain	YES	Concept foundation
S1-V2	LangChain Benefits	YES	Why use LangChain
S2-V1	What is OpenAI	YES	Quick watch
S2-V2	API Key Generation	MUST	■■■■■■ ■■■■■■■■ run ■■■■■!

Section 3 — Demo & Setup — Watch 1 of 2

Video	Topic	Action	Why Important
S3-V1	LangChain Demo	YES	Overview
S3-V2	Anaconda Installation	Use VS Code	

Section 4 — Models Module — WATCH (6 min only!)

Video	Topic	Action	Why Important
S4-V1	LangChain Modules Overview	YES	6 minutes — worth it!

Section 7 — Project 3: Embeddings — WATCH ALL 4 Videos

Video	Topic	Action	Why Important
S7-V1	Text Embedding Walkthrough	YES	Concept clear!
S7-V2	Embeddings Practical Python	YES	Run the code!
S7-V3	Embeddings Example Python	YES	Practice
S7-V4	Build Similar Words Finder App	+ BUILD	Mini Project!

Section 8 — Prompt Module — WATCH 4 of 5 Videos

Video	Topic	Action	Why Important
S8-V1	Prompts Module Introduction	YES	Overview
S8-V2	Prompt Template Walkthrough	YES	Daily use skill!
S8-V3	Example Selectors	YES	Advanced prompting
S8-V4	Adding More Examples	Basic	
S8-V5	Output Parsers Walkthrough	MUST	Clean JSON output!

Section 10 — Memory Module — WATCH ALL 2 Videos

Video	Topic	Action	Why Important
S10-V1	Importance of Memory in LLMs	MUST	Chatbot memory!
S10-V2	Different Types of Memory	MUST	ConversationMemory etc

Section 11 — ChatGPT Clone — WATCH ALL — BUILD PROJECT

Video	Topic	Action	Why Important
S11-V1	ChatGPT Clone Demo	YES	See end result first
S11-V2	Setting Up Project	YES	
S11-V3	Implementing Frontend	YES	

S11-V4	Modularizing Code	YES	Clean code practice
S11-V5	Passing Dynamic Data	YES	
S11-V6	Chatbot Conversational View	MUST	Key feature!
S11-V7	Conversation Summarization	MUST	Unique feature!

SKIP: Section 5 — Project 1 Simple Q&A; (C3 ■■ better ■■■■■■)

SKIP: Section 6 — Project 2 Conversational App (C3 ■■ already done)

SKIP: Section 9 — Project 4 Marketing Campaign (Outdated 2023 code)

COURSE 3 — Level Up LangChain + OpenAI (Step 4: Full Build)

Course:	Level Up LLM App Development with LangChain & OpenAI
Published:	November 2025 — Most practical & updated!
Total:	3h 51m — Watch ~3 hrs (skip MongoDB ~45 min)
Best Part:	Deploy ██████ Live URL ██████████ — Resume ███ Gold!

Section 1 — Introduction — WATCH (15 min)

Video	Topic	Action	Why Important
S1-V1	What You Should Know	YES	GitHub links save ██████

Section 2 — LangChain Basics — WATCH ALL 5 Videos

Video	Topic	Action	Why Important
S2-V1	Setup & Installation	YES	Environment ready
S2-V2	Create a Chain + LLM	MUST	Core concept!
S2-V3	Define & Structure Prompt	MUST	Important!
S2-V4	LCEL Syntax	MUST	2025 Standard!
S2-V5	Output Parsers	YES	Clean output

Section 3 — Similarity Search — WATCH ALL 4 Videos

Video	Topic	Action	Why Important
S3-V1	Installation & Setup	YES	
S3-V2	Create Embeddings from Text	MUST	Core RAG skill!
S3-V3	Querying Vector Store	YES	
S3-V4	Querying as Retriever	MUST	RAG pipeline piece

Section 4 — Full RAG Pipeline — WATCH ALL — BUILD PROJECT 1

Video	Topic	Action	Why Important
S4-V1	RAG Overview & Architecture	YES	Theory clear
S4-V2	Breaking Down RAG Pipeline	YES	
S4-V3	Project Setup	YES	
S4-V4	Load & Split Documents	MUST	PDF Chatbot ███ ██████████
S4-V5	Vector Store Chroma + Ingest	MUST	
S4-V6	Chain = Prompt+Model+Parser	MUST	
S4-V7	Chain + Retriever	MUST	
S4-V8	RunnablePassthrough	YES	
S4-V10	Solution — Chat History	MUST	Memory feature
S4-V11	Context + History Chatbot	+ BUILD	Your PROJECT 1!

Section 5 — Streamlit Web App — WATCH 5 of 6 Videos

Video	Topic	Action	Why Important
S5-V1	Setup Streamlit App	YES	
S5-V2	Build Layout	YES	
S5-V3	Adding Functionality	YES	
S5-V4	Challenge	Try yourself	
S5-V5	Add App to GitHub	MUST	Portfolio step!

S5-V6	Deploy App	MUST	Live URL get!
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Section 6 — Multi-Source Q&A; — WATCH 5 of 6 Videos

Video	Topic	Action	Why Important
S6-V1	Retrieval Query Analysis	YES	Concept
S6-V2	Connect Data Source + Index	YES	
S6-V3	Query Analysis Setup	YES	
S6-V4	Retrieval with Query Analysis	YES	
S6-V5	Challenge	Try yourself	
S6-V6	Solution Q&A; Multi-Source	+ BUILD	Your PROJECT 2!

Sections 7+8 — MongoDB — SKIP ENTIRELY

Video	Topic	Action	Why Important
S7+S8	MongoDB Vector Search + NoSQL	All videos	MongoDB specific — not needed

Section 9 — Tool Calling Agent — WATCH 5 of 7 Videos

Video	Topic	Action	Why Important
S9-V1	Agents to Perform Actions	YES	
S9-V2	Define Tools	MUST	Important!
S9-V3	Select Perfect Prompt	YES	Quick watch
S9-V4	Bind Tools & Create Agent	MUST	
S9-V5	Create & Run Agent Executor	MUST	
S9-V6	Challenge	Try yourself	
S9-V7	Solution — Tools & Functions	+ BUILD	Your PROJECT 3!

Section 10 — LangServe API — WATCH 4 of 6 Videos

Video	Topic	Action	Why Important
S10-V1	LangServe Installation	YES	
S10-V2	Create Server	YES	
S10-V3	Create Routes & Endpoints	MUST	REST API skill
S10-V4	Combine Prompt+Model+Output	YES	
S10-V5	Challenge	Try yourself	
S10-V6	Solution — Deploy RESTful API	MUST	

Section 11 — Cloud Deploy — WATCH ALL 3 Videos

Video	Topic	Action	Why Important
S11-V1	Manage & Deploy on Render	MUST	Live URL!
S11-V2	Create GitHub Repo + Push	MUST	Portfolio step!
S11-V3	Deploy Web Service	MUST	DONE!

COURSE 1 — AI Agent Dev: LangChain, Ollama, MCP (Step 5)

Course:	Master AI Agent Development — LangChain, OpenAI, Ollama, MCP
Published:	October 2025 — MCP is 2026 trending tech!
Total:	5h 6m — Watch ~2.5 hrs (skip ~2.5 hrs)
Key Skill:	FastMCP — Most searched AI skill in 2026!

Section 1 — Introduction to MCP — WATCH ALL 3 Videos — PRIORITY!

Video	Topic	Action	Why Important
S1-V1	What is MCP (with PPT)	MUST	2026 Most Trending Tech!
S1-V2	Understanding FastMCP	MUST	Python MCP Framework
S1-V3	FastMCP + Database + Ollama Agent	+ BUILD	Your PROJECT 4! (server.py)

Section 2 — Ollama + Local LLMs + RAG — WATCH 3 of 4 Videos

Video	Topic	Action	Why Important
S2-V1	Understanding Embeddings (PPT)	MUST	Deep embedding concept
S2-V2	First Local LLM — Qwen Model	YES	Run locally! (app.py)
S2-V3	RAG + LLaMA 3.1 + Pinecone	+ BUILD	Your PROJECT 5! (llama-rag.ipynb)
S2-V4	Deepseek Coder Assistant	Optional	Time ██████████

Section 3 — LangChain + OpenAI Agents — WATCH 3 of 6 Videos

Video	Topic	Action	Why Important
S3-V1	LangChain Agents + agent.ipynb	MUST	Agents deep dive
S3-V2	AI Voice Agent Emotion Analysis	Optional	Time ██████
S3-V3	Virtual Talking Agents	Optional	
S3-V4	AI Copilot Agent	+ BUILD	Good portfolio piece
S3-V5	Voice-to-GPT Agent	YES	Whisper + TTS cool!
S3-V6	AI Fitness & Diet Planner	Optional	

Section 4 — HuggingFace & GPT Models — SKIP ENTIRE SECTION

Video	Topic	Action	Why Important
S4 All	HuggingFace, GPT-2, Transformers	All 4 videos	Job ███ immediately ██████████

Section 5 — PyTorch — SKIP ENTIRE SECTION

Video	Topic	Action	Why Important
S5-V1	PyTorch Fundamentals	Not needed now	After job ██████████████████

Section 6 — LlamaIndex — Optional Watch

Video	Topic	Action	Why Important
S6-V1	What is LlamaIndex & Use Cases	OPTIONAL	Time █████ — 1 video only

Section 7 — Fine-Tuning — Watch 1 of 3 Only!

Video	Topic	Action	Why Important
S7-V1	Fine-Tuning Types & Techniques	Too deep	
S7-V2	Fine-Tuning VS RAG	MUST	Interview question!
S7-V3	Hands-On Fine-Tuning	Time waste	

Section 8 — Gemma Deploy — SKIP ENTIRE SECTION

Video	Topic	Action	Why Important
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S8-V1	Deploy Gemma on Colab + Ngrok	Complex setup	Not worth time now
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YOUR 5 PORTFOLIO PROJECTS — Job Ready!

Project 1	Context-Aware RAG Chatbot	Tech: LangChain + Chroma + OpenAI + Streamlit	Resume: "Built Context-Aware RAG Chatbot with Chat History"
Source: C3 — Section 4 + Section 5 Push to GitHub after building!			
Project 2	Multi-Source Q&A; Agent	Tech: LangChain + Query Analysis + Multiple Data Sources	Resume: "Built Multi-Data Source Q&A; Agent with Query Routing"
Source: C3 — Section 6 Push to GitHub after building!			
Project 3	Tool Calling AI Agent	Tech: LangChain + OpenAI Tools + LangServe + Render Deploy	Resume: "Built Autonomous Tool-Calling Agent — Deployed to Cloud"
Source: C3 — Section 9 + 10 + 11 Push to GitHub after building!			
Project 4	Enterprise MCP Data Agent	Tech: FastMCP + SQL Database + Ollama + Local LLM	Resume: "Built Enterprise Autonomous Data Agent with MCP Protocol"
Source: C1 — Section 1 (server.py) Push to GitHub after building!			
Project 5	Private RAG System (Capstone)	Tech: LLaMA 3.1 + Pinecone + Ollama + Hybrid RAG + Agentic RAG	Resume: "Built Privacy-First RAG System with Agentic Capabilities"
Source: C1 Section 2 + C5 Section 7,9 Push to GitHub after building!			

GITHUB + RESUME TIPS

1. ■■■■■■ Project ■■ README.md ■■■■■■ — Tech Stack, How to Run, Screenshots

2. ■■■■ GitHub ■■ Push ■■■■■■ — Green squares = Commitment proof

3. LinkedIn ■■ ■■■■■■ Project post ■■■■■■ — Visibility ■■■■■■■■■■

4. Resume ■■: Project Name | Tech Stack | GitHub Link | Live Demo Link

5. Profile Photo + About Section update ■■■■■■ LinkedIn ■■

6. Week 3 ■■■■■■ Job Apply ■■■■■■ — Wait ■■■■■■!

DAY-BY-DAY SCHEDULE (Based on Your Time)

1 Hour/Day — 56 Days

2 Hours/Day — 28 Days

4 Hours/Day — 14 Days (Fast Track)

Detailed Plan — 2 Hours/Day (28 Days)

Days	Step	Course + Section	Goal
Day 1-3	STEP 1	C2: Week 14 Python Basics	Python logic activate
Day 4-5	STEP 1	C2: Week 15 Intermediate Python	File/Dict/Exception
Day 6-7	STEP 1	C2: Week 16 APIs + JSON	API calls ready
Day 8	STEP 2	C5: Section 1 — What is RAG	RAG theory clear
Day 9	STEP 2	C5: Section 3 — Embeddings + Vector DB	Embedding concept
Day 10	STEP 3	C4: Sections 1,2,3,4 — LangChain Intro	LangChain basics
Day 11	STEP 3	C4: Section 7 — Embeddings Project	Similar Words App build
Day 12	STEP 3	C4: Section 8 — Prompt Module	Prompt Templates
Day 13	STEP 4	C3: Section 2 — LangChain + LCEL	Chains create
Day 14	STEP 4	C3: Section 3 — Similarity Search	Vector retrieval
Day 15-16	STEP 4	C3: Section 4 — Full RAG Pipeline	PROJECT 1 Start
Day 17	STEP 4	C3: Section 5 — Streamlit Deploy	PROJECT 1 Live!
Day 18	STEP 4	C3: Section 6 — Multi-Source Q&A;	PROJECT 2 Build
Day 19	STEP 4	C3: Section 9 — Tool Calling Agent	PROJECT 3 Start
Day 20	STEP 4	C3: Sections 10+11 — API + Cloud Deploy	PROJECT 3 Live!
Day 21	STEP 5	C1: Section 1 — MCP + FastMCP	PROJECT 4 Build
Day 22	STEP 5	C1: Section 2 — Ollama + RAG	PROJECT 5 Start
Day 23	STEP 5	C1: Section 3 — LangChain Agents	Agents deep
Day 24	STEP 6	C5: Section 4 — LangChain RAG Advanced	Metadata + Context
Day 25	STEP 6	C5: Section 5 — RAG Performance	Optimize + Metrics
Day 26	STEP 6	C5: Section 7 — Hybrid + Agentic RAG	PROJECT 5 upgrade!
Day 27	STEP 7	C4: Section 10 — Memory Module	Memory chatbot
Day 27	STEP 7	C4: Section 11 — ChatGPT Clone	Clone build
Day 28	STEP 8	C5: Section 9 — Capstone Project	Final PROJECT done!

JOBS YOU CAN APPLY — After Completing Roadmap

Job Title	Companies	Skills Required	Salary (India)
GenAI Engineer	TCS, Infosys, Wipro, Cognizant	LangChain, RAG, Python, LLM APIs	Rs 8-20 LPA
AI/LLM Developer	Startups, Product Companies	LangChain, OpenAI, Vector DB, Agents	Rs 12-25 LPA
RAG Engineer	GCCs, AI Companies	RAG, Embeddings, Pinecone, Chroma	Rs 15-35 LPA
MCP/AI Agent Dev	Startups, Freelance	FastMCP, Ollama, Agentic AI	Rs 15-30 LPA
Remote AI Engineer	US/UK Companies (Remote)	All above skills + Cloud Deploy	USD 40-80K/yr

IMPORTANT TIPS FOR SUCCESS

- Tip 1 — Code Along:** Videos ██████████ Code ████████. Copy-paste ████████. Typing ████████.
- Tip 2 — GitHub Daily:** ██████ ██████████████ GitHub ███ push ████████. Green squares = Dedication.
- Tip 3 — Don't Wait:** Week 3 ████████ Job Apply ██████████. Perfect ██████████ apply ██████ thinking ████████!
- Tip 4 — LinkedIn:** ████████ Project complete ██████████████████ LinkedIn ███ post ██████████ — 200+ views ██████████.
- Tip 5 — Stuck ████████:** 30 min stuck ████████ skip ██████ ██████████ ██████████. Later clear ██████████.
- Tip 6 — Interview Prep:** Fine-Tuning vs RAG, What is MCP, Explain RAG Pipeline — practice ██████████.
- Tip 7 — Projects First:** Videos ████████ Projects ██████████. ██████████ project build ██████████ real learning.

You have 5 Courses + 5 Projects + GitHub + LinkedIn That is EVERYTHING you need to land a Rs 8-15 LPA AI Job! Start Today. Be Consistent. Apply in Week 3. You've Got This!